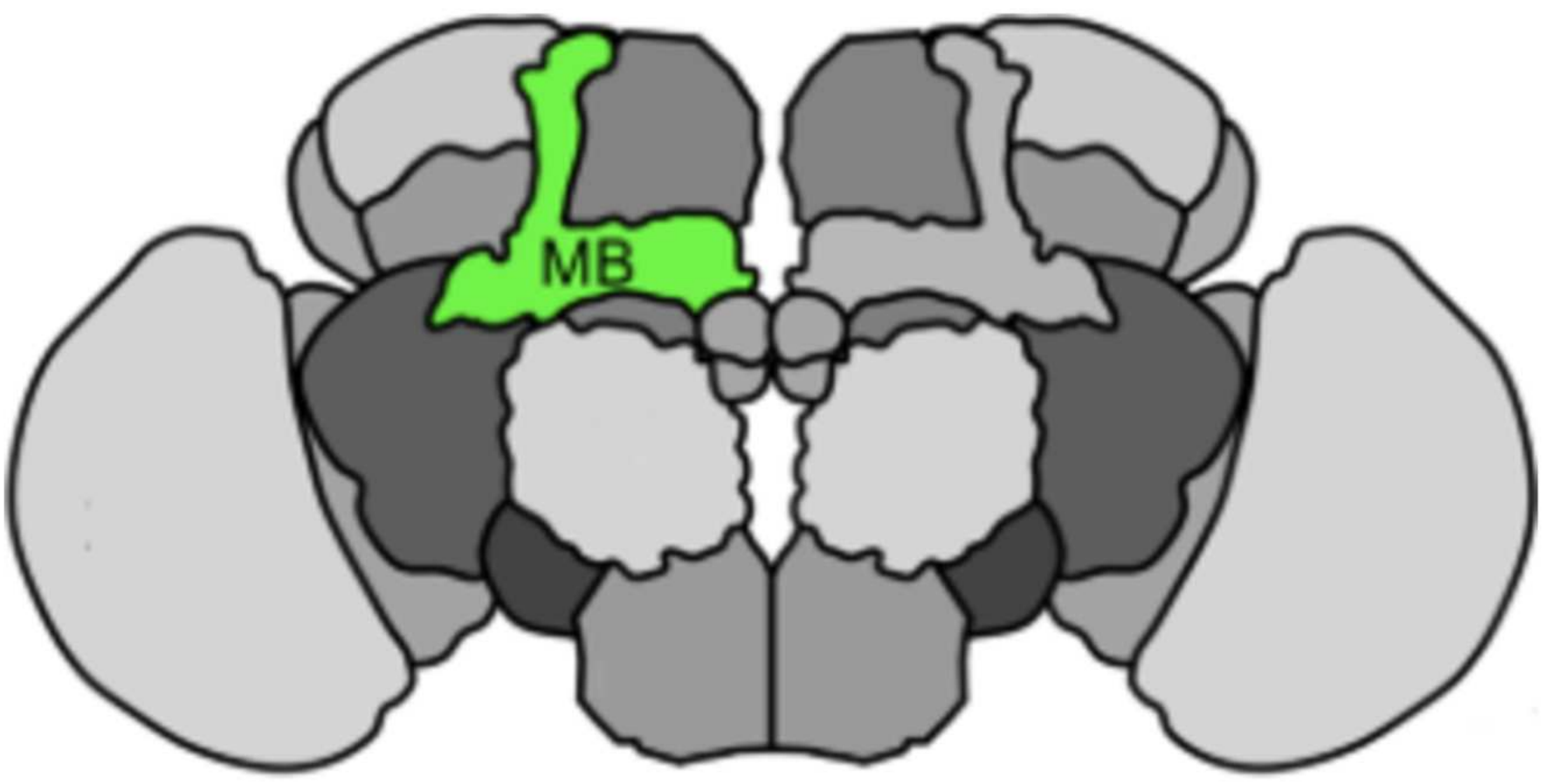


Introduction

- Can a brain with ~200,000 neurons monitor its own uncertainty? Metacognition—the ability to **evaluate decisions** under uncertainty—has been well characterized in primates and is strongly linked to **higher cortical regions** (Beran, 2019; Fleming & Lau, 2014; Garcia-Cordero et al., 2021).
- Assumption: **uncertainty monitoring** requires large, **complex brains**.
- Compact** nervous systems may also **support** advanced decision policies (Barron & Mourmourakis, 2023).
- Drosophila melanogaster* exhibits **complex** learning and behavioral flexibility despite its small brain (Rivi et al., 2023).
- Mediated in part by the **mushroom body (MB)**, a central integrative structure involved in **sensory** evaluation and **action** selection (Aso et al., 2014).

Figure 1: Mushroom Body Neurons in *Drosophila* (Bennett et al., 2021)



- Although flies have been extensively studied in associative learning paradigms, it remains **unclear** whether **flies** exhibit behavioral markers of **uncertainty monitoring** and whether mushroom body circuits contribute to these processes.
- Addressing this question tests whether metacognitive-like computations require **large cortical systems** or can emerge from **compact** neural circuitry.

Purpose

Research Question: Do flies exhibit behavioral markers of uncertainty monitoring, and does mushroom body activity modulate these behaviors?

Hypothesis: As visual uncertainty increases (lower contrast), wild-type flies will exhibit longer decision and checking latencies and higher abort rates. Silencing mushroom body neurons will disrupt these uncertainty-dependent behavioral adjustments without impairing baseline locomotion.

IV	Visual Contrast Condition (Uncertainty Level): Easy (High Contrast) / Medium / Hard (Low Contrast)
DVs	1. Baseline Locomotion Assay: Climbing Latency (s), Climb Success (0/1), Climbing Height (cm) 2. Contrast-Uncertainty Assay: Decision Latency (s), Checking Latency (s), Abort Rate (0/1), Success Rate (0/1).
Controls	Wild-type Flies and Parental Genotypes (29BD-Gal4, Kir2.1-UAS)

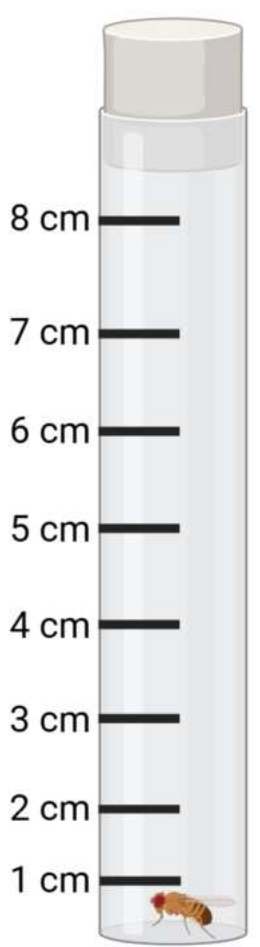
Metacognitive decision making and uncertainty monitoring in *Drosophila melanogaster*

Experimental Design

1. Baseline Locomotion Assay

Baseline locomotion was assessed for each genotype using a negative geotaxis assay to confirm intact motor function.

- Climb Latency (s): Time from tap-down to crossing the 8 cm mark.
- Success: Passing the 8 cm mark.
- Climbing Height (cm): Height each fly reaches at the end of the trial.



Genotypes

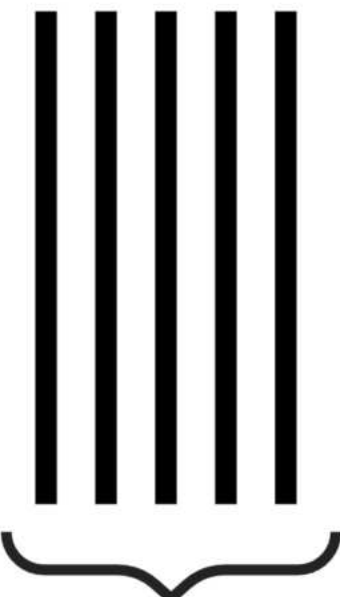
- WT
- 29BD-GAL4
- UAS-Kir2.1
- 29BD > Kir2.1 (MB-silenced)

Sample

- Random
- 5 Cohorts / Genotype
- n = 10 / Cohort
- 2 Trials per Test

2. Contrast-Uncertainty Assay

Visual Contrast Manipulation



Easy Uncertainty (High Contrast)

Medium

Hard Uncertainty (Low Contrast)

Contrast sleeves (created in InkSpace), were printed and placed around 75 % of the vial's circumference. Behavior was recorded during a negative geotaxis assay.

- Decision Latency (s): Time it takes to pass the 3 cm mark.
- Checking Latency (s): Time spent below 3 cm before sustained upward climbing begins.
- Abort: Never passing the 3 cm mark.
- Success: Passing the 8 cm mark.

Results

1. Contrast-Uncertainty Assay Results

Figure 2: Decision Latency Increases with Uncertainty (Created in JASP)

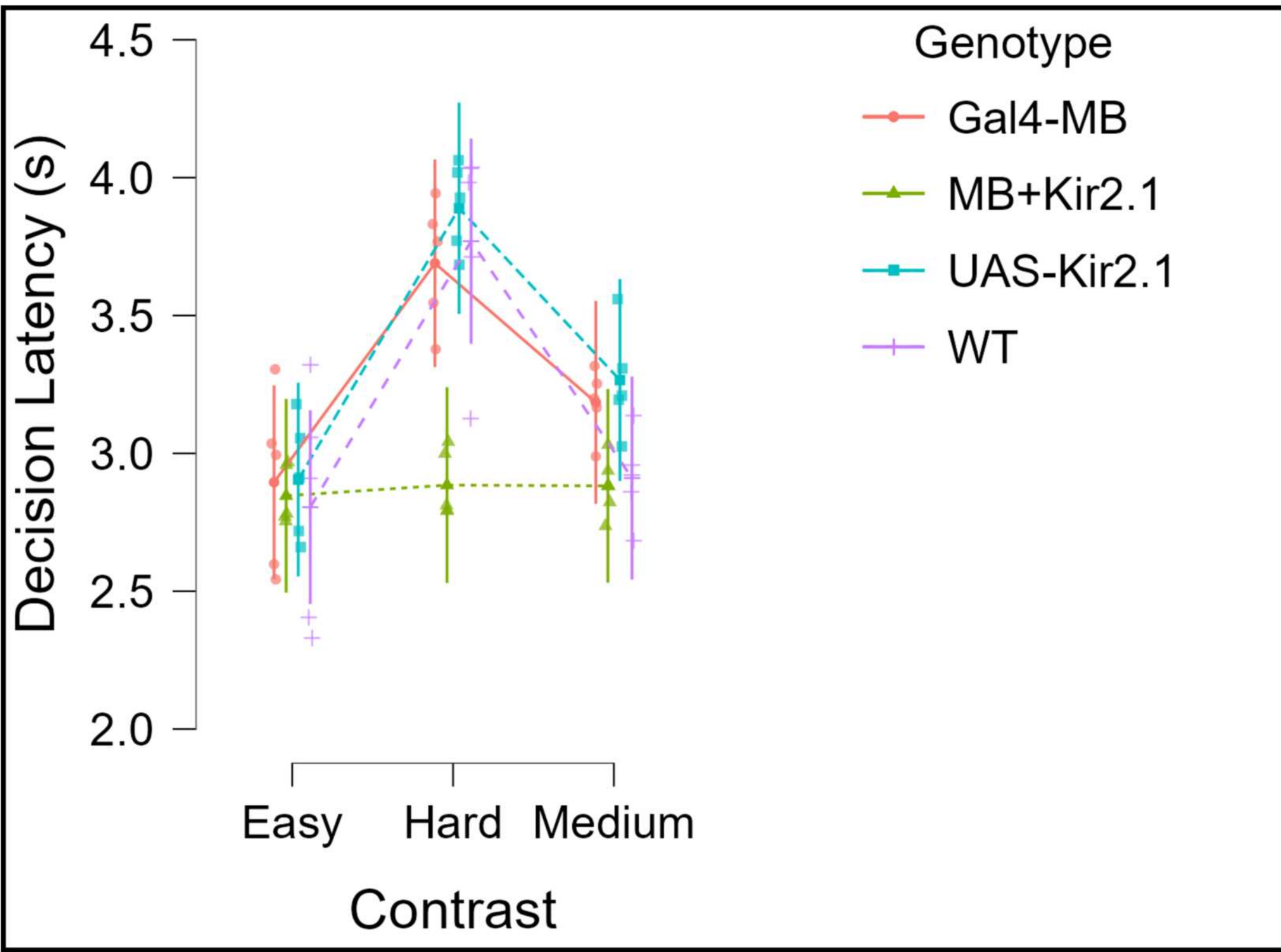
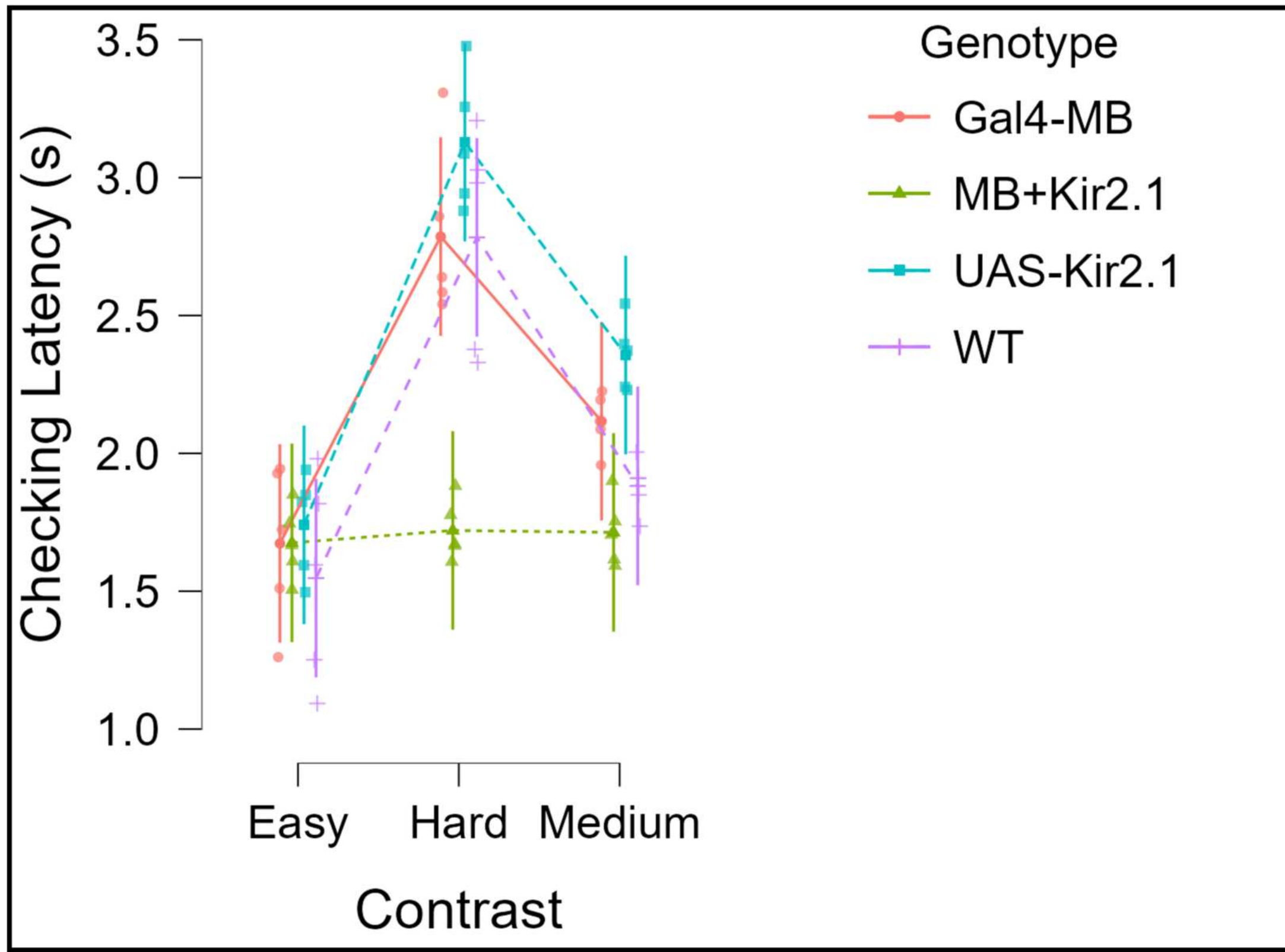


Figure 3: Checking Latency Increases with Uncertainty (Created in JASP)



2. Baseline Locomotion Assay Results

Figure 4: Climbing Latency Stays the Same Across Genotypes (Created in JASP)

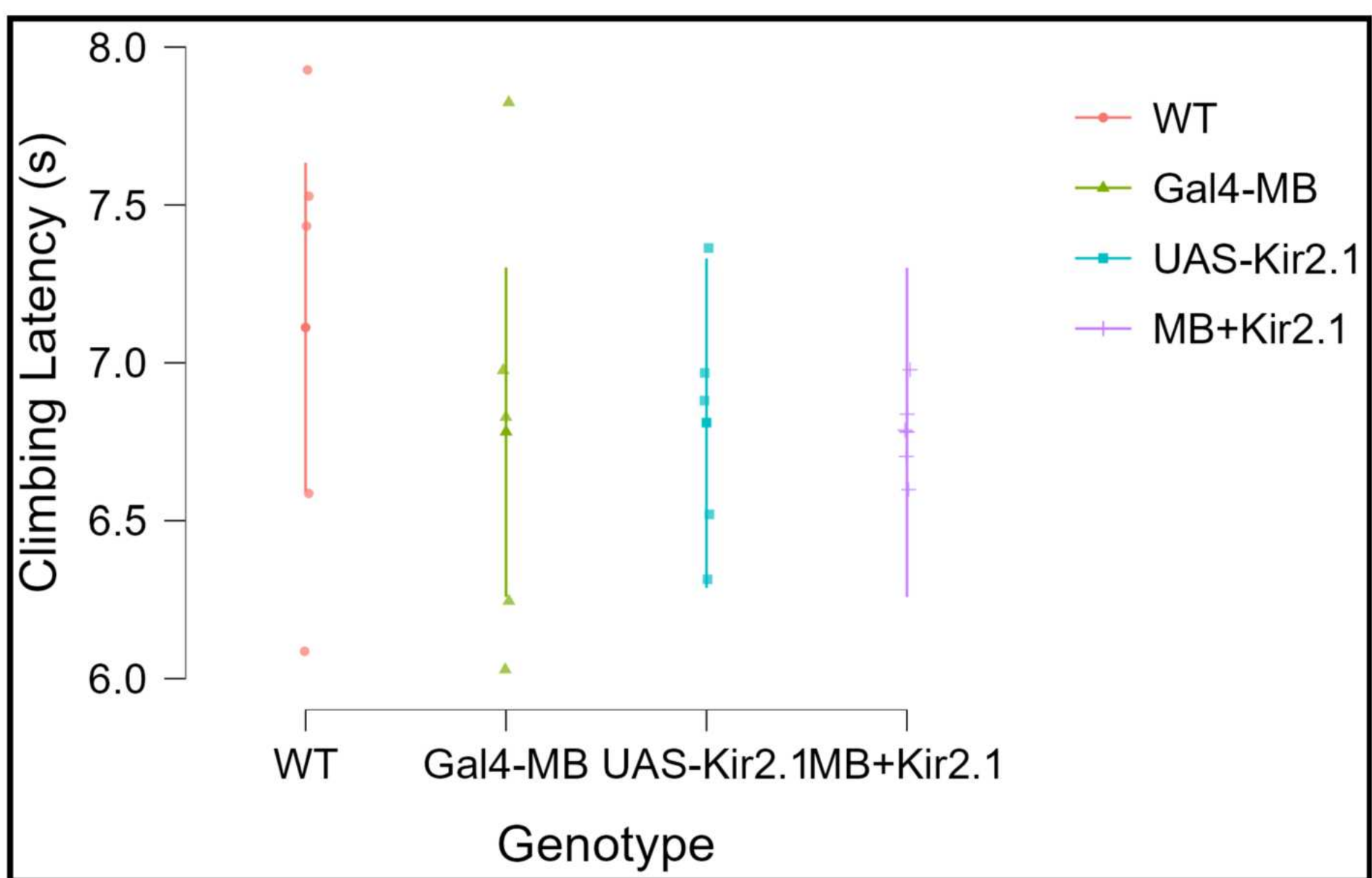
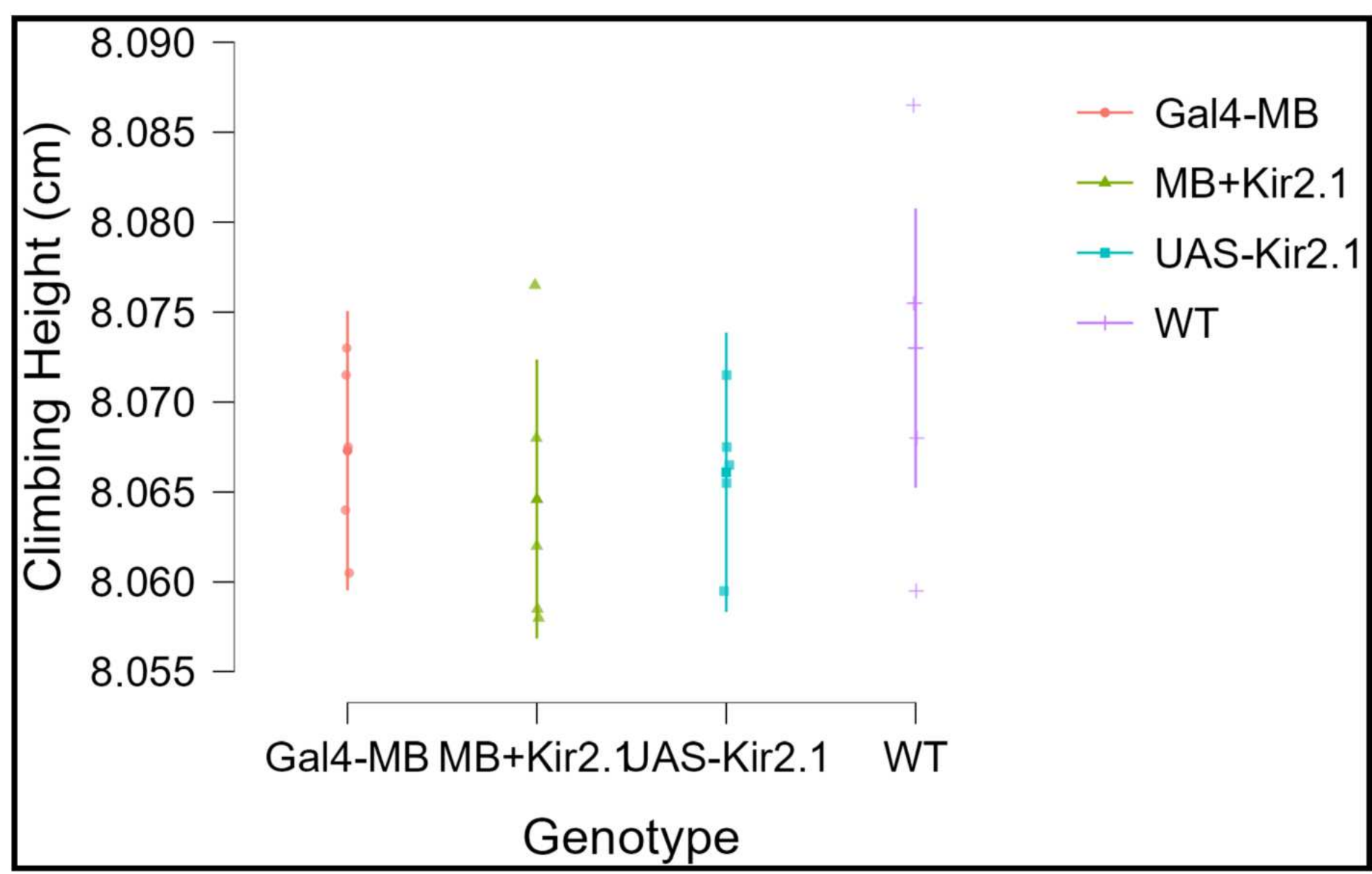


Figure 5: Climbing Height Stays the Same Across Genotypes (Created in JASP)



Tests Used: Linear mixed-effects models were used for climbing latency, climbing height, decision latency, and checking latency; generalized linear mixed models were used for abort and success rates.

Statistics

- Baseline Locomotion:** No genotype difference.
 - Climbing Latency:** $F(3,392) = 0.667$; $p = .573$.
 - Climbing Height:** $F(3,392) = 1.17$, $p = .319$.
 - Success Rate:** 100% success.
- Decision and Checking Latency:** Increased with visual uncertainty
 - Contrast:** $(F(2,1120) = 14.95, p < .001; F(2,1184) = 28.09, p < .001)$
 - Genotype:** $(F(3,1120) = 3.94, p = .008; F(3,1184) = 8.08, p < .001)$.
 - Genotype $\times$ Contrast (Decision Latency):** $(F(6,1120) = 1.77, p = .102)$, whereas
 - Genotype $\times$ Contrast (Checking Latency):** $(F(6,1184) = 3.03, p = .006)$. Reduced modulation of checking behavior under high uncertainty in mushroom body-silenced flies.
- Abort and Success Probabilities:**
 - Contrast:** $(\chi^2(2) = 34.85, p < .001; \chi^2(2) = 108.81, p < .001)$.
 - Genotype:** $(\chi^2(3) \approx 0, p = 1.00)$
 - Genotype $\times$ Contrast:** $(\chi^2(6) = 2.49, p = .869; \chi^2(6) = 0.27, p = 1.00)$.

Conclusion

- Flies adjusted behavior in response to visual uncertainty, showing **increased** decision and checking **latencies** as contrast decreased. **Silencing mushroom body neurons selectively disrupted uncertainty-dependent changes** in checking behavior while leaving locomotion intact. These findings suggest that mushroom body circuits **contribute** to evaluating uncertain **sensory** information. More broadly, the results support the idea that metacognitive-like decision processes can **emerge** from **compact** neural circuits.
- Future work combining neural recordings or targeted circuit manipulations could further reveal how uncertainty is encoded in the fly brain and how these mechanisms relate to neural dysfunction in neurological disorders.

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